Using the zoo

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1 Overview

EcologicalCellularAutomata collects published cellular automata for plant diseases and pests. Each model is a CellularAutomataCore `CAModel`, so it runs with the core's schemes, backends, callbacks and ensembles. Each also carries metadata: its source, what the source compared it with, and how faithfully it follows the source. This vignette shows how to find a model, read what it is and is not, build it, and run it.

```
ENV["GKSwstype"] = "100"
using EcologicalCellularAutomata
using CellularAutomataCore
using Plots
using Statistics
```

2 Finding a model

```
list_models()
```

```
26-element Vector{Symbol}:
 :bark_beetle_forest
 :barrier_island
 :bt_cotton_refuge
 :bt_crop_three_layer
 :cassava_mealybug
 :coffee_ant_fungus
 :coffee_rust
 :cvc_citrus
 :domany_kinzel_viral
 :dune_formation
 ⋮
 :pine_wilt_casim
 :potato_moth_farmers
 :resistance_regional
 :rhizoctonia_contact
 :rootworm_rotation
 :sudden_oak_death
 :telenomus_release
 :tuta_absoluta_asia
 :western_x_orchard
```

The models can be filtered by category or by fidelity:

```
(diseases=list_models(; category="plant disease"),
 pests=list_models(; category="agricultural pest"),
 reproductions=list_models(; fidelity=:reproduction))
```

```
(diseases = [:coffee_rust, :cvc_citrus, :domany_kinzel_viral, :fungus_dispersal,
:grapevine_leafroll, :rhizoctonia_contact, :sudden_oak_death, :western_x_orchard],
pests = [:bt_cotton_refuge, :bt_crop_three_layer, :cassava_mealybug,
:coffee_ant_fungus, :huffaker_mites, :ipydisp_dispersal, :potato_moth_farmers,
:resistance_regional, :rootworm_rotation, :telenomus_release, :tuta_absoluta_asia],
reproductions = [:domany_kinzel_viral, :eab_management, :emerald_ash_borer,
:fungus_dispersal, :ipydisp_dispersal, :mpb_multiscale, :potato_moth_farmers,
:rhizoctonia_contact, :rootworm_rotation, :telenomus_release])
```

3 What a model is, and is not

`model_info` returns the metadata of a model:

```
info = model_info(:fungus_dispersal)
```

```
fungus_dispersal: Fungus dispersal among permissive and resistant hosts
  plant disease: a hypothetical phytopathogenic fungus on a crop of permissive and
  resistant hosts
  source: Ferreira, I. E. de P., Moral, R. de A., Ferreira, C. P. and Godoy, W. A.
  C. (2013). Modelling fungus dispersal scenarios using cellular automata.
  Ecological Informatics 14, 53-58. https://doi.org/10.1016/j.ecoinf.2012.11.005
  fidelity: reproduction
  reproduces: Figure 3 (infected hosts over 120 days for all-permissive, mixed and
  all-resistant fields) and Figure 1 (infected hosts at day 120 against the daily
  propagule load).
  schemes: Synchronous
  deviations: 6
```

The fidelity says how close the model is to its source. A *reproduction* is the source's model with the source's parameters. An *adaptation* fills gaps in the source, or departs from it, and lists every such choice. An *inspired* model takes only the source's ideas. Even a reproduction makes choices where its source is silent:

```
foreach(d -> println("- ", d), info.deviations)
```

```
- The boundary is not stated in the source; the field is finite with nothing beyond
its edges (no wrap-around), so propagules leaving it are lost.
- The latency convention is not stated; a host infected in the step from day t first
disperses in the step from day t + latency. A latency of 0 is treated as 1, since a
synchronous step cannot infect and disperse at once.
- The nine initial infectious hosts form a 3 x 3 block at the centre; the source
says only 'at its centre'.
- The seasonal load is real-valued (not rounded) and evaluated at day t + 1 for the
step from time t.
- The source does not say how the 50% mixed field of its Figure 3 was laid out; the
random layout draws each host independently.
- Bands are vertical and start with a permissive band at the first column; the
source gives the width (15) but not the orientation.
```

A table of the whole zoo:

```
for id in list_models()
    m = model_info(id)
    println(rpad(id, 22), rpad(m.fidelity, 14), m.organism)
end
```

```
bark_beetle_forest    adaptation    spruce bark beetle
barrier_island        adaptation    dune-building grasses (burial-tolerant) and
woody shrubs (burial-intolerant)
bt_cotton_refuge      adaptation    Spodoptera frugiperda and Helicoverpa armigera
bt_crop_three_layer   adaptation    cotton bollworm (target pest), a mirid
(secondary pest) and a natural enemy
```

```
cassava_mealybug      adaptation    the cassava mealybug and the green lacewing as
its biological control
coffee_ant_fungus     adaptation    the ant Azteca instabilis, its scale insect
Coccus viridis and the fungus Lecanicillium lecanii
coffee_rust           adaptation    Hemileia vastatrix (coffee leaf rust)
cvc_citrus            adaptation    Xylella fastidiosa (citrus variegated
chlorosis), transmitted by sharpshooter leafhoppers
domany_kinzel_viral   reproduction  a lytic virus with heritable replicative ability
dune_formation        adaptation    dune-building plants (pioneer and stabilizing
species)
eab_management        reproduction  emerald ash borer (Agrilus planipennis)
emerald_ash_borer     reproduction  emerald ash borer (Agrilus planipennis)
fungus_dispersal      reproduction  a hypothetical phytopathogenic fungus
grapevine_leafroll    adaptation    grapevine leafroll-associated viruses, spread by
mealybugs
huffaker_mites        adaptation    the six-spotted mite Eotetranychus sexmaculatus
and the predatory mite Typhlodromus occidentalis
ipydisp_dispersal     reproduction  biological control agents (parasitoids and
predators of crop pests)
mpb_multiscale        reproduction  mountain pine beetle (Dendroctonus ponderosae)
pine_wilt_casim       adaptation    the pine wood nematode Bursaphelenchus
xylophilus, carried by Monochamus alternatus
potato_moth_farmers   reproduction  Andean potato tuber moth (Tecia solanivora)
resistance_regional   adaptation    a key insect pest with a single resistance locus
(generic)
rhizoctonia_contact   reproduction  Rhizoctonia solani (damping-off)
rootworm_rotation     reproduction  western corn rootworm (Diabrotica virgifera
virgifera)
sudden_oak_death      adaptation    Phytophthora ramorum
telenomus_release     reproduction  the neotropical brown stink bug Euschistus heros
and its egg parasitoid Telenomus podisi
tuta_absoluta_asia    adaptation    tomato leafminer (Tuta absoluta)
western_x_orchard     adaptation    Western X phytoplasma (Candidatus Phytoplasma
pruni), spread by leafhoppers
```

4 Building and running a model

`build` returns the model, with the source's parameter values as defaults:

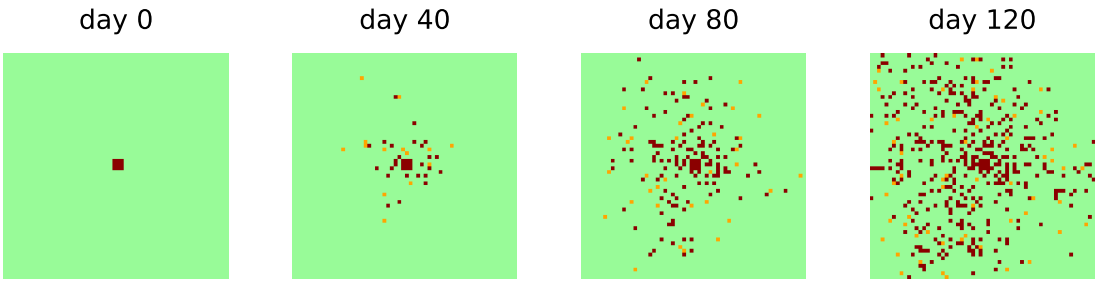
```
model = build(:fungus_dispersal)
```

```
CAModel fungus_dispersal
  Ferreira et al. (2013) fungus dispersal, range 3.0, seasonality none
  states:
    host::Int8 = 0 ∈ 0:2  # 0 healthy, 1 latent, 2 infectious
    resistant::Bool = false  # resistant (true) or permissive
    clock::Int16 = 0  # days since infection
  parameters:
    F_permissive = 0.003  # permissiveness of a permissive host
    F_resistant = 0.001  # permissiveness of a resistant host
```

```
    latency = 10  # latent period (days)
    load = 40.0  # daily propagule load (seasonality :none)
    load_max = 270.0  # peak daily load (seasonal)
    peak = 60.0  # day of peak load (:temperate)
    width = 10.0  # width of the season in days (:temperate)
    amplitude = 0.15  # relative amplitude (:tropical)
    period = 20.0  # period in days (:tropical)
  fields: infectious
  rule: LocalRule(<anonymous function>, VonNeumann(1))
```

`problem` returns a problem with the source's setup: here a 60×60 field with nine infectious hosts in the middle, for 120 days. Keywords change the setup, and `pmap` changes parameters by name:

```
prob = problem(:fungus_dispersal; seed=1)
sol = solve(prob; saveat=40)
plot([heatmap(u.host'; c=[:palegreen, :orange, :darkred], clims=(0, 2),
colorbar=false,
              aspect_ratio=:equal, framestyle=:none, title="day $t")
      for (t, u) in zip(sol.t, sol.u)]...; layout=(1, 4), size=(900, 250))
```



Healthy hosts are green, latent hosts orange and infectious hosts dark red. Changing a parameter, here halving the daily propagule load:

```
half = solve(problem(:fungus_dispersal; seed=1, pmap=(; load=20.0));
save_everystep=false)
(full_load=fungus_infected(sol.u[end]), half_load=fungus_infected(half.u[end]))
```

```
(full_load = 446, half_load = 193)
```

5 Replicates

The models are stochastic, and a single run says little. Most problem constructors draw their initial state from `seed` as well, so replicates are problems built with different seeds. Here the infected hosts at day 120 over ten replicates:

```
finals = [fungus_infected(solve(problem(:fungus_dispersal; seed=s);
                                save_everystep=false).u[end]) for s in 1:10]
(mean=mean(finals), sd=round(std(finals); digits=1))
```

```
(mean = 471.9, sd = 43.7)
```

The core's `EnsembleProblem` runs replicates in parallel. Its `prob_func` builds each replicate's problem from SciMLBase's ensemble context, whose `sim_id` numbers the replicates:

```
ens = EnsembleProblem(problem(:rhizoctonia_contact);
                      prob_func=(prob, ctx) -> problem(:rhizoctonia_contact;
seed=ctx.sim_id))
runs = solve(ens, Synchronous(), EnsembleThreads(); trajectories=50,
save_everystep=false)
mean(rhizoctonia_infected(r.u[end]) for r in runs.u)
```

```
20.46
```

6 Any backend

A model is specified once. The core's backends run it, and those that support it agree exactly, because every random number comes from a counter-based generator keyed by the seed, the cell, the time and a draw number. The viral front model on three backends:

```
prob = problem(:domany_kinzel_viral; L=2000, tspan=200, seed=3)
threaded = solve(prob; backend=ArrayBackend(), save_everystep=false).u[end]
serial = solve(prob; backend=ArrayBackend(; threaded=false),
save_everystep=false).u[end]
reference = solve(prob; backend=ReferenceBackend(), save_everystep=false).u[end]
threaded == serial == reference
```

```
true
```

7 Models with callbacks

Some sources describe steps that no local rule can express. Mites in the Huffaker model act one at a time, and their moves write two cells. Insecticide in the Bt cotton model is sprayed when either pest reaches a share of the whole field. Such parts are callbacks stored with the problem, so `solve` runs them without being told:

```
prob = problem(:bt_cotton_refuge; n=200, initial=50, scenario=:insecticide)
keys(prob.kwargs)
```

```
(:callback,)
```

They draw their random numbers from a generator seeded by the problem's seed and the time, so a run is reproducible:

```
a = solve(prob; save_everystep=false).u[end]
b = solve(prob; save_everystep=false).u[end]
a == b
```

```
true
```

The following vignettes take the models one at a time. Each compares the model with its source's results and repeats its deviations.